

Keenan Heinz

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EDUCATION

University of Pennsylvania

BSE, Mechanical Engineering, Concentration in Structures & Materials

August 2024 – May 2028

Philadelphia, PA

ENGINEERING EXPERIENCE

Mechanical Engineering Intern

Stealth Startup

June 2026 – Present

Menlo Park, CA

Automated Test Rig Fleet (Mini-Instron DAQ Systems)

- **Requirements & Structural Design:** Set load cases and specimen requirements per rig; ran calc-driven fixture and fastener designs (bolted-joint preload, bending, deflection, fatigue) to a 2.0 minimum yield safety factor.
- **Electrical, Controls & DAQ Software (Working Knowledge):** Applied first-principles circuit and motor-control theory, with AI-assisted tools, to build wiring/control stacks (STM32, ADC, load cells) and SPI-based DAQ software for synchronized force-displacement-time capture; self-taught DFM for outsourced fabrication.

Non-Contact Optical Extensometer

- **Optics, Vision & Sync:** Selected lens and working-distance parameters and used AI-assisted tools to develop vision-processing software and camera-controller synchronization for non-contact strain measurement and post-processing.

Coil Winding System (ITC-PLC-70)

- **Mechanical Design:** Designing custom bobbins and a tensioning system for an ITC-PLC-70 coil winding machine.
- **Software & Validation:** Building HMI-integrated software to set coil parameters; running test winds to validate scalability for future production.

Co-Lead Engineer

September 2025 – Present

Penn Hyperloop

Philadelphia, PA

- **Mechanical Design:** Engineered, in Fusion 360, an A36 steel propulsion structure for an in-house Tunnel Boring Machine (TBM), rated for vertical loads >2500 N and horizontal loads >5000 N.
- **Structural Validation:** Ran analytical calculations and Finite Element Analysis (FEA) to confirm component integrity under worst-case loading.
- **Actuator Sourcing:** Evaluated 50+ suppliers to redesign the actuation system, hitting 150 kN thrust and 1+ m stroke targets while minimizing cost.
- **Team Leadership:** Promoted from Propulsion Engineer to Co-Lead Engineer; now directs mechanical-electrical integration across sub-teams for the 2027 machine.

SKILLS

- **CAD & Simulation:** SolidWorks, Fusion 360, Onshape, FEA, DraftSight, MATLAB
- **Electrical & Controls:** wiring, signal chains, STM32, stepper drivers, ADCs, load cells (working knowledge, AI-assisted)
- **Manufacturing:** DFM, production engineering (self-taught), MIG welding, ironworker, band saws, drill press, miter saw, belt sanders, hand tools
- **Programming:** Python, Java, Excel
- **Design:** DaVinci Resolve, CapCut, Photoshop, Pixlr, PowerPoint, Canva
- **Interests:** backpacking and camping, road biking, classic rally homologation cars, rock music, cats

ATHLETICS

D1 Student Athlete, Men's Heavyweight Crew

August 2024 – Present

University of Pennsylvania

- Two-time National Championship qualifier: 8th, 2026 IRA National Championship; 9th, 2025 IRA National Championship (team best points finish).

National Team Athlete, USRowing National Team

June 2023 – August 2024

- 2nd, 2024 World Rowing Championships (St. Catharine's); 5th, 2023 World Rowing Championships (Paris).